#Linux #github

https://github.com/antonaleks/LinuxPractice

Введение

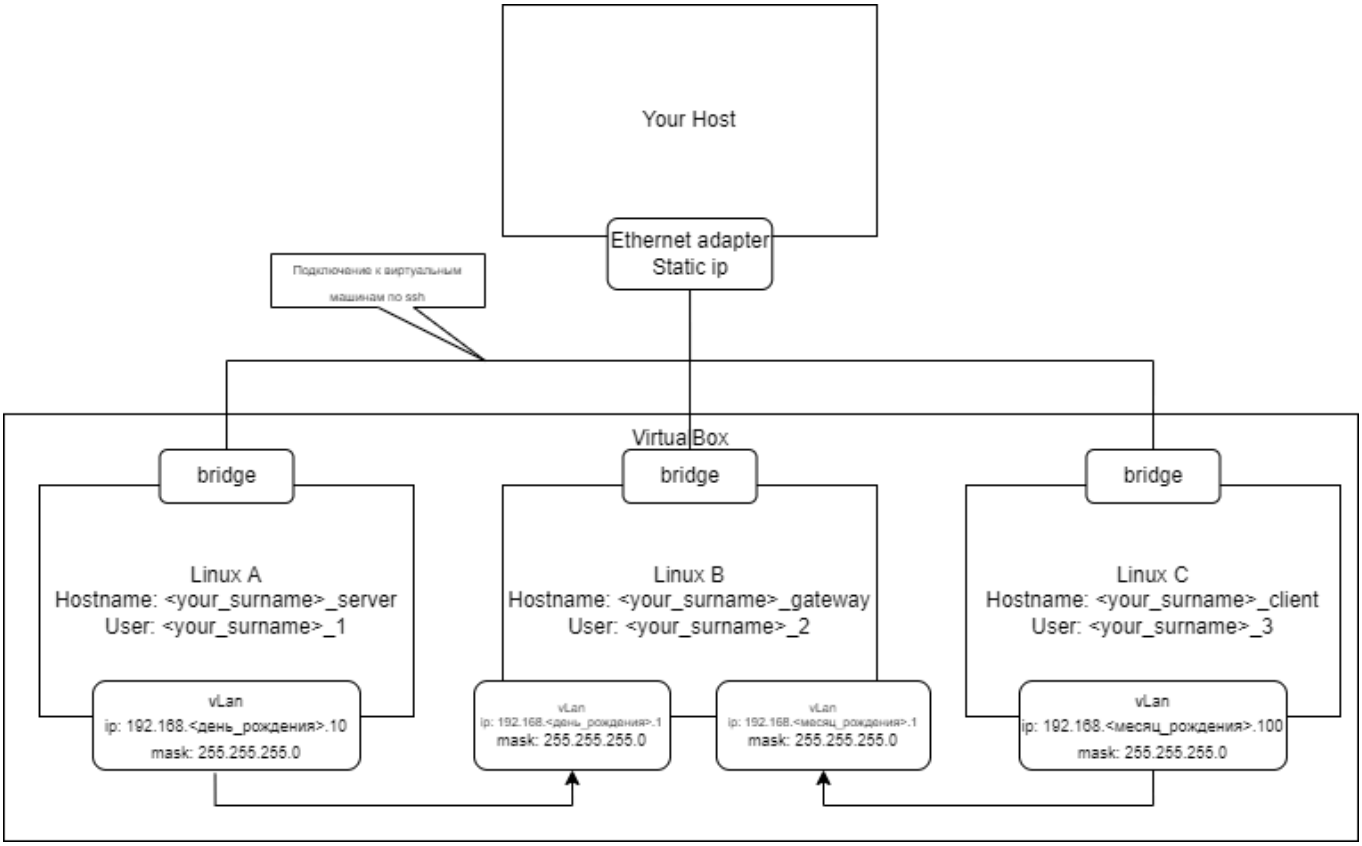
Данная практика посвещена базовым принципам администрирования Linux. Для выполнения практики необходимо развернуть 3 виртуальные машины Linux.

Как пример, можно использовать Virtual Box + Ubuntu Server 20.04.

Задание

Для выполнения данного задания необходимо:

1. Развернуть три виртуальные машины Linux, согласно схеме ниже



2. Linux A

- [x] ~~Сконфигурировать Hostname следующим образом: <your_surname>_server (пропустить если делаете через Play with docker)~~
- [x] ~~Создать пользователя <your_surname>_1 (пропустить если делаете через Play with docker)~~

- [x] ~~Сконфигурировать виртуальный интерфейс со следующим ip адресом: 192.168.<день рождения>.10/24~~
- [x] ~~Развернуть Http сервер на виртуальной машине на порту 5000. Необходимо реализовать минимум три эндпоинта (запрос /get, /post, /put)~~

2. Linux B

- [x] ~~Сконфигурировать Hostname следующим образом: <your_surname>_gateway (пропустить если делаете через Play with docker)~~
- [x] ~~Создать пользователя <your_surname>_2 (пропустить если делаете через Play with docker)~~
- [x] ~~Сконфигурировать 2 виртуальных интерфейс со следующими ip адресом: 192.168.<день рождения>.1/24, 192.168.<месяц рождения>.10/24~~
- [x] ~~С помощью утилит ip route и iptables настроить маршрут пакетов от Linux A до C. Должны быть запрещены все пакеты, кроме http пакетов через порт 5000 (маршруты обязательно, файрвол опционально)~~
- [x] ~~Запустить программу tcpdump с фильтрацией по портам 5000~~

2. Linux C

- [x] ~~Сконфигурировать Hostname следующим образом: <your_surname>_client (пропустить если делаете через Play with docker)~~
- [x] ~~Создать пользователя <your_surname>_3 (пропустить если делаете через Play with docker)~~
- [x] ~~Сконфигурировать виртуальный интерфейс со следующим ip адресом: 192.168.<месяц рождения>.100/24~~
- [x] ~~С помощью команды curl на машине C послать 3 запроса на машину A в http сервер (/get, /post, /put)~~
- [x] ~~При перезагрузки системы все сервисы и сетевая архитектура должны также функционировать (сохранить свои настройки)~~
- [x] ~~Сделать скриншоты всех этапов задания~~
- [x] ~~Оформить отчет в виде Markdown файла. Приложить конфигурационные файлы в репозиторий~~
- [x] ~~Для play with docker также нужно написать bash скрипты для воспроизведения на новых вм~~

Репозиторий оформить следующим образом:

- application - исходный код http сервера
- configs - конфигурационные файлы сетевых настроек виртуальных машин
- report.md - Отчет markdown

P.S. Если не получилось использовать разные утилиты для настройки сети, используйте netplan Смотрите мануал - там информация для virtual box и Play with docker

Отчет по практической работе: Базовые принципы администрирования Linux

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Цель работы: Настройка трех виртуальных машин Linux (сервер, шлюз и клиент), конфигурация сети между ними, настройка маршрутизации с фильтрацией трафика и развертывание просто веб-сервера.

Инструменты: VirtualBox, Ubuntu Server 20.04.06 LTS.

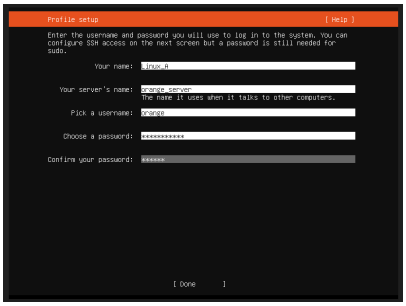
Данные:

- Фамиля: Абышев
- День рождения: 21
- Месяц рождени: май (5)

Глава 1. Подготовка виртуальных машин

Linux_A

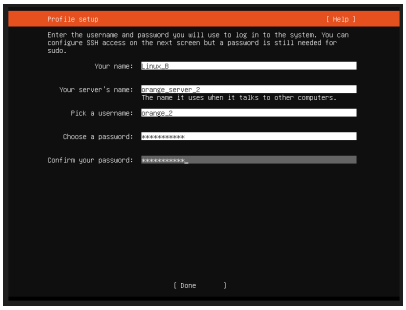
```
Your name: Linux_A
Your server's name: orange_server
Pick a username: orange
```



Linux_B

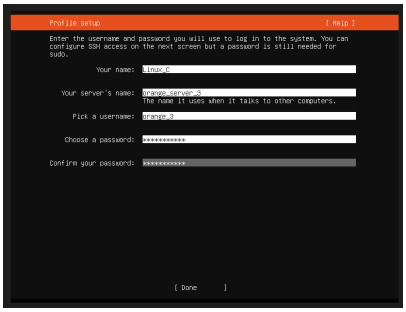
```
Your name: Linux_B
Your server's name: orange_server_2
```

```
Pick a username: orange_2
```



Linux_C

```
Your name: Linux_C
Your server's name: orange_server_3
Pick a username: orange_3
```



Глава 2. Настройка Linux A (сервер)

2.1. Имя хоста и пользователь

- Hostname: orange_server
- Пользователь: orange

  2.2. Настройка сети

  Сетевые адаптеры в VirtualBox:

  - Адаптер 1: Сетевой мост (для доступа)
  - Адаптер 2: Внутренняя сеть

Сгенерируем ключ:

```
Last login: Tue Mar 10 04:56:34 UTC 2026 on tty1
orange@orangeserver:~$ ssh-keygen -t rsa -N "" -f my.key
Generating public/private rsa key pair.
Your identification has been saved in my.key
Your public key has been saved in my.key.pub
The key fingerprint is:
SHA256:TLtyTnbC9IMmiufDLfVNGLILf2PFeBNrODMVzVFKuTO orange@orangeserver
The key's randomart image is:
+---[RSA 3072]----+
|           oo+.  |
|          ..+.   |
|         .  ..o  |
|       .o..o . E |
|        oSB o   . |
|     . ooB+O     |
|    . =ooXX+.    |
|    .=.+O=o..    |
|   .ooo o..      |
+----[SHA256]-----+
orange@orangeserver:~$
```

Подставим IP:

```
orange@orangeserver:~$ ssh-copy-id -i my.key.pub orange@192.168.31.160
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "my.key.pub"
The authenticity of host '192.168.31.160 (192.168.31.160)' can't be established.
ECDSA key fingerprint is SHA256:/auL6Mg6IrbgNzUuWw/L3nkb+OFA4zIHRZGMeEc3SkI.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are alr
eady installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to inst
all the new keys
orange@192.168.31.160's password:
Permission denied, please try again.
orange@192.168.31.160's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'orange@192.168.31.160'"
and check to make sure that only the key(s) you wanted were added.

orange@orangeserver:~$ _
```

Конфигурация Netplan (/etc/netplan/00-installer-config.yaml):

```
  GNU nano 4.8                    /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      addresses:
       - 192.168.21.10/24
      dhcp4: false
    enp0s8:
      dhcp4: true
  version: 2
```

Проверка IP - адресов после применения netplan:

```
orange@orangeserver:~$ ip route
default via 192.168.31.1 dev enp0s8 proto dhcp src 192.168.31.160 metric 100
192.168.21.0/24 dev enp0s3 proto kernel scope link src 192.168.21.10
192.168.31.0/24 dev enp0s8 proto kernel scope link src 192.168.31.160
192.168.31.1 dev enp0s8 proto dhcp scope link src 192.168.31.160 metric 100
orange@orangeserver:~$ _
```

2.3. Настройка веб-сервера (Flask):

Установим flask:

```
Setting up libgcc-9-dev:amd64 (9.4.0-1ubuntu1~20.04.2) ...
Setting up libexpat1-dev:amd64 (2.2.9-1ubuntu0.8) ...
Setting up libpython3.8-dev:amd64 (3.8.10-0ubuntu1~20.04.18) ...
Setting up zlib1g-dev:amd64 (1:1.2.11.dfsg-2ubuntu1.5) ...
Setting up cpp (4:9.3.0-1ubuntu2) ...
Setting up gcc-9 (9.4.0-1ubuntu1~20.04.2) ...
Setting up libpython3-dev:amd64 (3.8.2-0ubuntu2) ...
Setting up libstdc++-9-dev:amd64 (9.4.0-1ubuntu1~20.04.2) ...
Setting up gcc (4:9.3.0-1ubuntu2) ...
Setting up g++-9 (9.4.0-1ubuntu1~20.04.2) ...
Setting up python3.8-dev (3.8.10-0ubuntu1~20.04.18) ...
Setting up g++ (4:9.3.0-1ubuntu2) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
Setting up build-essential (12.8ubuntu1.1) ...
Setting up python3-dev (3.8.2-0ubuntu2) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for mime-support (3.64ubuntu1) ...
orange@orangeserver:~$ pip3 install flask_
```

Завершена:

```
Processing triggers for mime-support (3.64ubuntu1) ...
orange@orangeserver:~$ pip3 install flask
Collecting flask
  Downloading flask-3.0.3-py3-none-any.whl (101 kB)
     |████████████████████████████████| 101 kB 5.1 MB/s
Collecting blinker>=1.6.2
  Downloading blinker-1.8.2-py3-none-any.whl (9.5 kB)
Collecting click>=8.1.3
  Downloading click-8.1.8-py3-none-any.whl (98 kB)
     |████████████████████████████████| 98 kB 7.8 MB/s
Collecting Jinja2>=3.1.2
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
     |████████████████████████████████| 134 kB 17.7 MB/s
Collecting itsdangerous>=2.1.2
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting Werkzeug>=3.0.0
  Downloading werkzeug-3.0.6-py3-none-any.whl (227 kB)
     |████████████████████████████████| 227 kB 8.6 MB/s
Collecting importlib-metadata>=3.6.0; python_version < "3.10"
  Downloading importlib_metadata-8.5.0-py3-none-any.whl (26 kB)
Collecting MarkupSafe>=2.0
  Downloading MarkupSafe-2.1.5-cp38-cp38-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (26 kB)
Collecting zipp>=3.20
  Downloading zipp-3.20.2-py3-none-any.whl (9.2 kB)
Installing collected packages: blinker, click, MarkupSafe, Jinja2, itsdangerous, Werkzeug, zipp, imp
ortlib-metadata, flask
  WARNING: The script flask is installed in '/home/orange/.local/bin' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-s
cript-location.
Successfully installed Jinja2-3.1.6 MarkupSafe-2.1.5 Werkzeug-3.0.6 blinker-1.8.2 click-8.1.8 flask-
3.0.3 importlib-metadata-8.5.0 itsdangerous-2.2.0 zipp-3.20.2
orange@orangeserver:~$
```

Создадим директорию для http_server:

```
orange@orangeserver:~$ sudo mkdir /opt/http_server
```

Проверим текущую директорию с помощью команды pwd и создадим файл orange_5000.py

```
orange@orangeserver:~$ pwd
/home/orange
orange@orangeserver:~$ cd /opt/http_server
orange@orangeserver:/opt/http_server$ sudo nano orange_5000.py
```

```
  GNU nano 4.8                                orange_5000.py
from flask import  Flask, request, jsonify

app = Flask(name)

@app.route('/get', "methods"=['GET'])
def get():
        return jsonify({"method": "GET", "message": "Get response"}), 200

@app.route('/post', "methods"=['POST'])
def post():
        data = request.get_json()
        return jsonify("method": "POST", "received": data), 200

@app.route('/put', "methods"=['PUT'])
def put():
        data = request.get_json()
        return jsonify({"method": "PUT", "received":data}),200

if name == main
        app.run(host='0.0.0.0', port=5000)
```

Проверим путь к файлу flask с помощью команды which flask:

```
orange@orangeserver:~$ which flask
/home/orange/.local/bin/flask
```

Создадим systemd unit файл, который будет запускать сервер автоматически при старте системы:

```
  GNU nano 4.8                  /etc/systemd/system/http_service.service
[Unit]
Description=HTTP_Server_5000
After=network.target

[Service]
User=orange
WorkingDirectory=/opt/http_server
ExecStart=/usr/bin/python3 /opt/http_server/orange_5000.py
Environment=/home/orange/.local/bin/flask run --host=0.0.0.0 --port=5000
Restart=always

[Install]
WantedBy=multi-user.target
```

Включим и запустим:

```
orange@orangeserver:~$ sudo systemctl enable http_service.service
Created symlink /etc/systemd/system/multi-user.target.wants/http_service.service → /etc/systemd/syst
em/http_service.service.
orange@orangeserver:~$ sudo systemctl start http_service.service
orange@orangeserver:~$
```

Результат:

```
orange@orangeserver:/opt/http_server$ sudo systemctl restart http_server
orange@orangeserver:/opt/http_server$ sudo systemctl status http_server
● http_server.service - HTTP_Server_5000
     Loaded: loaded (/etc/systemd/system/http_server.service; enabled; vendor preset: enabled)
     Active: active (running) since Mon 2026-03-09 06:09:24 UTC; 4s ago
   Main PID: 1349 (python3)
      Tasks: 1 (limit: 4610)
     Memory: 17.9M
     CGroup: /system.slice/http_server.service
             └─1349 /usr/bin/python3 /opt/http_server/orange_5000.py

Mar 09 06:09:24 orangeserver systemd[1]: Started HTTP_Server_5000.
Mar 09 06:09:25 orangeserver python3[1349]:  * Serving Flask app 'orange_5000'
Mar 09 06:09:25 orangeserver python3[1349]:  * Debug mode: off
Mar 09 06:09:25 orangeserver python3[1349]: WARNING: This is a development server. Do not use it in>
Mar 09 06:09:25 orangeserver python3[1349]:  * Running on all addresses (0.0.0.0)
Mar 09 06:09:25 orangeserver python3[1349]:  * Running on http://127.0.0.1:5000
Mar 09 06:09:25 orangeserver python3[1349]:  * Running on http://127.0.0.1:5000
Mar 09 06:09:25 orangeserver python3[1349]: Press CTRL+C to quit
lines 1-17/17 (END)
```

C:\Users\Ormon Abyshev\OneDrive\Desktop

Глава 3. Настройка Linux B (шлюз)

3.1. Имя хоста и пользователь

- Hostname: orange_server_2
- Пользователь: orange_2

  3.2. Настройка сети:

  Сетевые адаптеры в VirtualBox:

  - Адаптер 1: Внутренняя сеть
  - Адаптер 2: Внутренняя сеть
  - Адаптер 3: Сетевой мост (для доступа)

Отредактируем файл net-plan:

```
orange_2@orangeserver2:~$ sudo nano /etc/netplan/00-installer-config.yaml
```

```
  GNU nano 4.8                    /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: true
  version: 2
```

Внесенные изменения с помощью команды netplan apply:

```
  GNU nano 4.8                    /etc/netplan/00-installer-config.yaml                    Modified
# This is the network config written by 'subiquity'
network:
  version: 2
  ethernets:
    enp0s3:
      addresses:
       - 192.168.21.1/24
      dhcp4: false
    enp0s8:
      addresses:
       - 192.168.5.10/24
      dhcp4: false
    enp0s9:
      dhcp4: true

File Name to Write: /etc/netplan/00-installer-config.yaml_
^G Get Help          M-D DOS Format       M-A Append           M-B Backup File
^C Cancel            M-M Mac Format       M-P Prepend          ^T To Files
```

Посмотрим изменения через ip a:

```
orange_2@orangeserver2:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:ce:2c:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.21.1/24 brd 192.168.21.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fece:2cd2/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:a4:57:e3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.5.10/24 brd 192.168.5.255 scope global enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fea4:57e3/64 scope link
       valid_lft forever preferred_lft forever
orange_2@orangeserver2:~$
```

3.3. Настройка маршрутизации и файрвола:

Включение IP Forwaarding:

```
  GNU nano 4.8                    /etc/sysctl.conf                    Modified
#
# /etc/sysctl.conf - Configuration file for setting system variables
# See /etc/sysctl.d/ for additional system variables.
# See sysctl.conf (5) for information.
#

#kernel.domainname = example.com

# Uncomment the following to stop low-level messages on console
#kernel.printk = 3 4 1 3

###############################################################3
# Functions previously found in netbase
#

# Uncomment the next two lines to enable Spoof protection (reverse-path filter)
# Turn on Source Address Verification in all interfaces to
# prevent some spoofing attacks
#net.ipv4.conf.default.rp_filter=1
#net.ipv4.conf.all.rp_filter=1

# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1

# Uncomment the next line to enable packet forwarding for IPv6
#  Enabling this option disables Stateless Address Autoconfiguration
#  based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1

^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos    M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Paste Text ^T To Spell   ^_ Go To Line M-E Redo
```

```
orange_2@orangeserver2:~$ sudo sysctl -p
net.ipv4.ip_forward = 1
orange_2@orangeserver2:~$
```

По умолчанию запрещаем весь пересылаемый через сервер трафик:

```
orange_2@orangeserver2:~$ sudo iptables -P FORWARD DROP
orange_2@orangeserver2:~$ _
```

Разрешим уже установленные ранее соединения:

```
orange_2@orangeserver2:~$ sudo iptables -A FORWARD -m state --state ESTABLISHED,RELATED -j ACCEPT
orange_2@orangeserver2:~$
```

Разрешим трафик между подсетями только для порта 5000 (весь остальной FORWARD будет дропаться по умолчанию)

```
orange_2@orangeserver2:~$ sudo iptables -A FORWARD -s 192.168.21.0/24 -d 192.168.5.0/24 -p tcp --dpo
rt 5000 -j ACCEPT
orange_2@orangeserver2:~$ sudo iptables -A FORWARD -s 192.168.5.0/24 -d 192.168.21.0/24 -p tcp --spo
rt 5000 -j ACCEPT
orange_2@orangeserver2:~$ _
```

Проверим внесенные правила с помощью iptables -L:

```
orange_2@orangeserver2:~$ sudo iptables -A FORWARD -s 192.168.21.0/24 -d 192.168.5.0/24 -p tcp --dpo
rt 5000 -j ACCEPT
orange_2@orangeserver2:~$ sudo iptables -A FORWARD -s 192.168.5.0/24 -d 192.168.21.0/24 -p tcp --spo
rt 5000 -j ACCEPT
orange_2@orangeserver2:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination

Chain FORWARD (policy DROP)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere             state RELATED,ESTABLISHED
ACCEPT     tcp  --  192.168.21.0/24      192.168.5.0/24       tcp dpt:5000
ACCEPT     tcp  --  192.168.5.0/24       192.168.21.0/24      tcp spt:5000

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
orange_2@orangeserver2:~$
```

Сохраним внесенные изменения:

```
Configuring iptables-persistent

Current iptables rules can be saved to the configuration file /etc/iptables/rules.v6. These
rules will then be loaded automatically during system startup.

Rules are only saved automatically during package installation. See the manual page of
ip6tables-save(8) for instructions on keeping the rules file up-to-date.

Save current IPv6 rules?

<Yes>                                <No>

Selecting previously unselected package netfilter-persistent.
(Reading database ... 72282 files and directories currently installed.)
Preparing to unpack .../netfilter-persistent_1.0.14ubuntu1_all.deb ...
Unpacking netfilter-persistent (1.0.14ubuntu1) ...
Selecting previously unselected package iptables-persistent.
Preparing to unpack .../iptables-persistent_1.0.14ubuntu1_all.deb ...
Unpacking iptables-persistent (1.0.14ubuntu1) ...
Setting up netfilter-persistent (1.0.14ubuntu1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/netfilter-persistent.service → /lib/syst
emd/system/netfilter-persistent.service.

Progress: [ 56%] [##############################################.................................]
```

```
orange_2@orangeserver2:~$ sudo sysctl -p
net.ipv4.ip_forward = 1
orange_2@orangeserver2:~$ sudo iptables -P FORWARD DROP
orange_2@orangeserver2:~$ sudo iptables -A FORWARD -m state --state ESTABLISHED,RELATED -j ACCEPT
orange_2@orangeserver2:~$ sudo iptables -A FORWARD -s 192.168.21.0/24 -d 192.168.5.0/24 -p tcp --dpo
rt 5000 -j ACCEPT
orange_2@orangeserver2:~$ sudo iptables -A FORWARD -s 192.168.5.0/24 -d 192.168.21.0/24 -p tcp --spo
rt 5000 -j ACCEPT
orange_2@orangeserver2:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination

Chain FORWARD (policy DROP)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere             state RELATED,ESTABLISHED
ACCEPT     tcp  --  192.168.21.0/24      192.168.5.0/24       tcp dpt:5000
ACCEPT     tcp  --  192.168.5.0/24       192.168.21.0/24      tcp spt:5000

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
orange_2@orangeserver2:~$ sudo netfilter-persistent save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables save
orange_2@orangeserver2:~$
```

Глава 4. Настройка Linux C

4.1. Имя хоста и пользователь

- Hostname: orange_server_3
- Пользователь: orange_3

  4.2. Настройка сети:

  Сетевые адаптеры в VirtualBox:

  - Адаптер 1: Внутренняя сеть
  - Адаптер 2: Сетевой мост (для доступа)

Проверим текущие интерфейсы:

```
orange_3@orangeserver3:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:15:b6:02 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::a00:27ff:fe15:b602/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:cc:31:05 brd ff:ff:ff:ff:ff:ff
    inet 192.168.31.174/24 brd 192.168.31.255 scope global dynamic enp0s8
       valid_lft 42869sec preferred_lft 42869sec
    inet6 fe80::a00:27ff:fecc:3105/64 scope link
       valid_lft forever preferred_lft forever
orange_3@orangeserver3:~$ _
```

Конфигурация Netplan (/etc/netplan/00-installer-config.yaml

```
  GNU nano 4.8                    /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      addresses:
      - 192.168.5.100/24
      routes:
       - to: default
         via: 192.168.5.10
      dhcp4: false
    enp0s8:
      dhcp4: true
  version: 2

                                      [ Read 13 lines ]
^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify     ^C Cur Pos     M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste Text  ^T To Spell    ^_ Go To Line  M-E Redo
```

Проверка IP- адрессов:

```
orange_3@orangeserver3:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:15:b6:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.5.100/24 brd 192.168.5.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe15:b602/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:cc:31:05 brd ff:ff:ff:ff:ff:ff
    inet 192.168.31.174/24 brd 192.168.31.255 scope global dynamic enp0s8
       valid_lft 35842sec preferred_lft 35842sec
    inet6 fe80::a00:27ff:fecc:3105/64 scope link
       valid_lft forever preferred_lft forever
orange_3@orangeserver3:~$ _
```

Установка curl:

```
orange_3@orangeserver3:~$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
curl is already the newest version (7.68.0-1ubuntu2.25).
curl set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
orange_3@orangeserver3:~$
```

Глава 5. Тестирование

5.1. Запуска tcpdump на шлюзе (Linux_B):

```
orange_2@orangeserver2:~$ sudo tcpdump -i any port 5000 -n
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on any, link-type LINUX_SLL (Linux cooked v1), capture size 262144 bytes
_
```

5.2. Отправка запросов с клиента Linux_C на сервер Linux_A

С клиента (C) выполнены команды curl к серверу A:

- GET:

```
orange_3@orangeserver3:~$ curl -X GET http://192.168.21.10:5000/get
_
```

```
orange_2@orangeserver2:~$ sudo tcpdump -i any port 5000 -n
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on any, link-type LINUX_SLL (Linux cooked v1), capture size 262144 bytes
04:08:22.485628 IP 192.168.5.100.36138 > 192.168.21.10.5000: Flags [S], seq 4038110460, win 64240, o
ptions [mss 1460,sackOK,TS val 2806960764 ecr 0,nop,wscale 7], length 0
04:08:23.487904 IP 192.168.5.100.36138 > 192.168.21.10.5000: Flags [S], seq 4038110460, win 64240, o
ptions [mss 1460,sackOK,TS val 2806961766 ecr 0,nop,wscale 7], length 0
04:08:25.503728 IP 192.168.5.100.36138 > 192.168.21.10.5000: Flags [S], seq 4038110460, win 64240, o
ptions [mss 1460,sackOK,TS val 2806963782 ecr 0,nop,wscale 7], length 0
```

- POST:

```
orange_3@orangeserver3:~$ curl -X POST http://192.168.21.10:5000/post -H "Content-Type: application/
json" -d '{"test":"data"}'
_
```

```
orange_2@orangeserver2:~$ sudo tcpdump -i any port 5000 -n
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on any, link-type LINUX_SLL (Linux cooked v1), capture size 262144 bytes
04:16:41.628097 IP 192.168.5.100.55964 > 192.168.21.10.5000: Flags [S], seq 313884124, win 64240, op
tions [mss 1460,sackOK,TS val 2807459910 ecr 0,nop,wscale 7], length 0
04:16:49.820229 IP 192.168.5.100.55964 > 192.168.21.10.5000: Flags [S], seq 313884124, win 64240, op
tions [mss 1460,sackOK,TS val 2807468102 ecr 0,nop,wscale 7], length 0
```

- PUT:

```
orange_3@orangeserver3:~$ curl -X PUT http://192.168.21.10:5000/put -H "Content-Type: application/js
on" -d '{"test":"data"}'
```

```
orange_2@orangeserver2:~$ sudo tcpdump -i any port 5000 -n
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on any, link-type LINUX_SLL (Linux cooked v1), capture size 262144 bytes
04:19:08.013595 IP 192.168.5.100.36452 > 192.168.21.10.5000: Flags [S], seq 3759982174, win 64240, o
ptions [mss 1460,sackOK,TS val 2807606296 ecr 0,nop,wscale 7], length 0
04:19:09.018904 IP 192.168.5.100.36452 > 192.168.21.10.5000: Flags [S], seq 3759982174, win 64240, o
ptions [mss 1460,sackOK,TS val 2807607302 ecr 0,nop,wscale 7], length 0
04:19:11.037050 IP 192.168.5.100.36452 > 192.168.21.10.5000: Flags [S], seq 3759982174, win 64240, o
ptions [mss 1460,sackOK,TS val 2807609319 ecr 0,nop,wscale 7], length 0
04:19:15.227918 IP 192.168.5.100.36452 > 192.168.21.10.5000: Flags [S], seq 3759982174, win 64240, o
ptions [mss 1460,sackOK,TS val 2807613510 ecr 0,nop,wscale 7], length 0
^C
```

5.4. Проверка блокировки другого трафика

Попытка выполнить ping с клиента C на сервер A была неуспешной (пакеты блокируются шлюзом B):

```
orange_3@orangeserver3:~$ ping 192.168.21.10
PING 192.168.21.10 (192.168.21.10) 56(84) bytes of data.
^C
--- 192.168.21.10 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2031ms

orange_3@orangeserver3:~$
```

Глава 6. Проверка сохранения настроек после перезагрузки

Все три виртуальные машины были перезагружены с помощью команды sudo reboot. После перезагрузки выполнены следующие проверки:

- IP Forwarding на Linux B активен:

```
 * Support:        https://ubuntu.com/advantage

  System information as of Tue 10 Mar 2026 04:24:58 AM UTC

  System load:  0.72              Users logged in:         0
  Usage of /:   31.5% of 8.02GB   IPv4 address for enp0s3: 192.168.21.1
  Memory usage: 5%                IPv4 address for enp0s8: 192.168.5.10
  Swap usage:   0%                IPv4 address for enp0s9: 192.168.31.247
  Processes:    118

 * Ubuntu 20.04 LTS Focal Fossa has reached its end of standard support
   on 31 May 2025.

   For more details see:
   https://ubuntu.com/20-04

 * Introducing Expanded Security Maintenance for Applications.
   Receive updates to over 25,000 software packages with your
   Ubuntu Pro subscription. Free for personal use.

     https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection
or proxy settings


Last login: Tue Mar 10 03:55:34 UTC 2026 on tty1
orange_2@orangeserver2:~$ cat /proc/sys/net/ipv4/ip_forward
1
orange_2@orangeserver2:~$ _
```

- Правила iptables на Linux_B загружены:

```
Last login: Tue Mar 10 03:55:34 UTC 2026 on tty1
orange_2@orangeserver2:~$ cat /proc/sys/net/ipv4/ip_forward
1
orange_2@orangeserver2:~$ sudo iptables -L FORWARD -v -n
[sudo] password for orange_2:
Chain FORWARD (policy DROP 60 packets, 3792 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  *      *       0.0.0.0/0            0.0.0.0/0            state RELAT
ED,ESTABLISHED
    0     0 ACCEPT     tcp  --  *      *       192.168.21.0/24      192.168.5.0/24       tcp dpt:500
0
    0     0 ACCEPT     tcp  --  *      *       192.168.5.0/24       192.168.21.0/24      tcp spt:500
0
orange_2@orangeserver2:~$
```

- Веб-сервер на Linux_A запущен:

```
   Receive updates to over 25,000 software packages with your
   Ubuntu Pro subscription. Free for personal use.

     https://ubuntu.com/pro

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection
or proxy settings


Last login: Tue Mar 10 03:26:08 UTC 2026 on tty1

orange@orangeserver:~$ systemctl status http_service.service
● http_service.service - HTTP_Server_5000
     Loaded: loaded (/etc/systemd/system/http_service.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2026-03-10 04:24:25 UTC; 6min ago
   Main PID: 652 (python3)
      Tasks: 1 (limit: 4610)
     Memory: 20.9M
     CGroup: /system.slice/http_service.service
             └─652 /usr/bin/python3 /opt/http_server/orange_5000.py

Mar 10 04:24:25 orangeserver systemd[1]: Started HTTP_Server_5000.
Mar 10 04:24:27 orangeserver python3[652]:  * Serving Flask app 'orange_5000'
Mar 10 04:24:27 orangeserver python3[652]:  * Debug mode: off
Mar 10 04:24:27 orangeserver python3[652]: WARNING: This is a development server. Do not use it in >
Mar 10 04:24:27 orangeserver python3[652]:  * Running on all addresses (0.0.0.0)
Mar 10 04:24:27 orangeserver python3[652]:  * Running on http://127.0.0.1:5000
Mar 10 04:24:27 orangeserver python3[652]:  * Running on http://192.168.31.160:5000
Mar 10 04:24:27 orangeserver python3[652]: Press CTRL+C to quit
lines 1-17/17 (END)
```

- Тестовый запрос с Linux_C на Linux_A:

```
orange_3@orangeserver3:~$ curl -X GET http://192.168.21.10:5000/get
_
```

```
orange_2@orangeserver2:~$ sudo tcpdump -i any port 5000 -n
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on any, link-type LINUX_SLL (Linux cooked v1), capture size 262144 bytes
04:32:24.456110 IP 192.168.5.100.52062 > 192.168.21.10.5000: Flags [S], seq 2155982363, win 64240, o
ptions [mss 1460,sackOK,TS val 526722595 ecr 0,nop,wscale 7], length 0
04:32:25.467886 IP 192.168.5.100.52062 > 192.168.21.10.5000: Flags [S], seq 2155982363, win 64240, o
ptions [mss 1460,sackOK,TS val 526723607 ecr 0,nop,wscale 7], length 0
04:32:26.812020 IP 192.168.5.100.56212 > 192.168.21.10.5000: Flags [S], seq 86102411, win 64240, opt
ions [mss 1460,sackOK,TS val 526724951 ecr 0,nop,wscale 7], length 0
04:32:27.484856 IP 192.168.5.100.52062 > 192.168.21.10.5000: Flags [S], seq 2155982363, win 64240, o
ptions [mss 1460,sackOK,TS val 526725623 ecr 0,nop,wscale 7], length 0
04:32:31.675978 IP 192.168.5.100.52062 > 192.168.21.10.5000: Flags [S], seq 2155982363, win 64240, o
ptions [mss 1460,sackOK,TS val 526729815 ecr 0,nop,wscale 7], length 0
_
```

```
orange@orangeserver:~$ curl http://localhost:5000/get
{"message":"Get response","method":"GET"}
orange@orangeserver:~$ _
```

Глава 7. Вывод

В ходе выполнения практической работы №1 "Базовые принципы администрирования Linux" была успешна настроена сетевая инфраструктура из трех виртуальных машин Linux. Настроен IP - адрес для сервера (A) и клиента (C), а также два статических IP для

шлюза (B). С помощью iptables на шлюзе настроена маршрутизация и фильтрация трафика, разрешающая только HTTP - запросы на порт 5000 от клиента к серверу. Веб - сервер FLask был развернут на сервере и настроен для автозапуска с помощью systemd. Все настройки сохраняются после перезагрузки системы.

Используемая литература:

1. https://1cloud.ru/help/linux/change_ip_in_ubuntu_20 - # Замена IP в Ubuntu 20.04 с помощью net-tools.
2. https://habr.com/ru/articles/783574/ - # Flask для начинающих
3. https://flask.palletsprojects.com/en/stable/quickstart/
4. https://habr.com/ru/companies/slurm/articles/255845/ - # Systemd за пять минут
5. https://tokmakov.msk.ru/blog/item/494
6. https://timeweb.cloud/docs/unix-guides/iptables-setup
7. https://habr.com/ru/articles/747616/ - # Введение в Iptables
8. https://pc.ru/articles/poluchenie-root-prav-pri-rabote-v-winscp - # Получение root прав при работе в WinSCP